

Wentao Yang

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Education

Pennsylvania State University

Aug. 2026

Ph.D. in Statistics (Machine Learning Track)

University of Chicago

Sep. 2024 – Mar. 2026

M.S. in Computational and Applied Mathematics

Cumulative GPA: 3.8/4.0 with Graduated Honors Thesis

Israel Institute of Technology (Technion)

Sep. 2020 – Jun. 2024

B.Sc Mathematics with Computer Science

Cumulative GPA: 94/100

Honors & Awards

Graduate Fellowship (Full Funding), Pennsylvania State University

2026

Student Excellence Award, Pennsylvania State University

2026

Academic Excellence Scholarship (Rank top 5%), Technion

2020 – 2024

Dean's List, Technion

2020 – 2024

Research Experience

Online Planning Monte-Carlo Algorithm for Partially Observable Markov Decision Processes Problems (POMDP)

May 2025 – Mar 2026

UChicago, Master Thesis Advisor: Prof. Mihai Anitescu

Chicago, IL

- Designed the belief-state filter update algorithm and the heuristic-based rollout policy for new nodes in search-down exploration.
- Completed the proof that the estimated value function converges to the optimal value function as the number of simulations increases, and conducted experiments on the Rock Sample problem to validate the result. And successfully defended the thesis.

Cooperative of Conditional Flow Matching (CFM) and Decoder for Reconstruction From Low-dimensional Latent Vectors

Apr. 2025 – Mar 2026

UChicago, Supervisor: Prof. Nisha Chandramoorthy

Chicago, IL

- Trained a cooperative of CFM and decoder models, where CFM generates low-dimensional feature vectors, and decoder reconstructs synthesized images.
- Designing and testing various network architectures and training loss functions to validate this approach and identify the best-fit feature vectors across different datasets.

Accelerating Abnormal-Behavior Detection in Surveillance Video Via Deep Neural Networks and Group Testing Methods

Aug. 2023 – Dec. 2023

Computer Science Department, Technion, Advisor: Dr. Catherine Haddad

Haifa, Israel

- Designed the architecture of convolutional neural networks (CNNs) and sample preprocessing.

Construct Various LR Parsing Tables with Test Estimation and Optimization of Parsing Algorithms

Feb. 2023 – Jun. 2023

GTIT, Supervisor: Prof. Guillermo Morales-Luna

Shantou, China

- Lowered the space-consuming when constructing the table and adjusted the core of sets of LR(1) to solve parsing-action conflicts by syntax and lexical analysis.

Internship Experience

Developer of Machine Vision R&D Department

Jun. 2023 – Aug. 2023

Shenzhen Huahan Weiye Tech Co. Ltd, Director: Dr. Yang Yang

Shenzhen, China

- Optimized the backpropagation algorithm to improve the detection accuracy rate. Constructed and trained Convolutional Neural Networks (CNN) to detect defects on the production line with colleagues.

Teaching Experience

Teaching Assistant

Sep. 2023 - May. 2024

Mathematics with Computer Science Department

Haifa, Israel

- Served as a tutor for Dr. Aviram's course on Infinitesimal Calculus 1, aiding in student comprehension and coursework.

GO Teacher

Jul. 2020 - Sep. 2020

Nanjing Qingyuan Go School

Nanjing, China

- Taught high-level go-player students and helped hold competitions with almost 500 go-players.

Extracurricular Experience

Course Assistant

Sep. 2022 - Jun. 2024

Teaching Faculty

Haifa, Israel

- Assisted Prof. Guillermo Morales-Luna in grading Data Structure and Theory of Computation homework, and designed the syllabus of courses.

Peer Tutor

Sep. 2021 - Jun. 2024

Engineering and Mathematics with Computer Science Departments

Shantou, China

- Taught students how to learn math and helped them with their study problems.

Peer Mentor

Aug. 2021 - Oct. 2021

Cohort 2021 Faculty

Shantou, China

- Organized ice-breaking and other social events for freshmen to build networks.
- Provided support for freshmen to adapt themselves to university life.

Language & Skills

Language: Mandarin (native) and English (Fluent)

Coding: Python, C language, C++ language, and Java

Go: amateur 5-dan rank